

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: CSA51 COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO5: *Develop the network security strategies based on the study of viruses, worms, firewall architectures, and IDS/IPS functionalities.CO (BL5, BL6)*

Assessment Tool Weightage: 40%

QUESTIONS: 5 TOTAL MARKS: 50 Annexure A
CONSTRAINT-BASED PROBLEM SOLVING

Code of Conduct:

*I **T Sanhith(192311228)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Annexure A

CONSTRAINT-BASED PROBLEM SOLVING

1. A financial institution must secure all web transactions using TLS, but legacy systems only support older SSL versions. Design a solution to ensure secure communication.

Constraints:

- A. Backward compatibility must be maintained for 20% legacy clients**
- B. Sensitive data must not be transmitted using SSL**
- C. System downtime must be minimal during migration**
- D. Cost of implementation should be optimized**

Problem Context: Secure web transactions via TLS while maintaining legacy SSL support for 20% of clients, keeping sensitive data off SSL, ensuring minimal downtime, and optimizing cost.

Proposed Solution: Reverse Proxy / Application Delivery Controller (ADC) Routing

Implement a cost-effective reverse proxy (like NGINX or HAProxy) configured to act as an SSL termination point and traffic router.

Constraint	How the Solution Addresses It
A. Backward compatibility	The reverse proxy allows older protocols (e.g., TLS 1.0/SSLv3) <i>only</i> for specific routing paths, successfully handshaking with the 20% legacy clients.
B. No sensitive data on SSL	WAF (Web Application Firewall) rules on the proxy inspect the negotiated protocol. If the client connects via SSL, the proxy blocks access to sensitive URI paths (e.g., <code>/checkout</code> , <code>/login</code>) and redirects them to an "Upgrade Browser" page.
C. Minimal downtime	By deploying the proxy in parallel with the current infrastructure, you can execute a Blue-Green deployment. DNS is switched to the proxy, meaning zero downtime for the actual

	application servers.
D. Optimized cost	Using enterprise-grade open-source tools like HAProxy or NGINX eliminates the need for expensive proprietary hardware load balancers.

2. An e-commerce platform is considering implementing the SET protocol for payment security but currently uses TLS-based encryption. Analyze and propose a decision.

Constraints:

A. Must support high transaction throughput (10,000 transactions/minute)

B. Compatibility with modern browsers is required

C. Implementation complexity should be minimized

D. Regulatory compliance (PCI-DSS) must be ensured

Problem Context: Evaluating the SET (Secure Electronic Transaction) protocol versus TLS for high-throughput (10k/min), modern browser-compatible, low-complexity, and PCI-DSS compliant e-commerce transactions.

Decision: Reject SET and upgrade to Modern TLS (1.2/1.3) + Tokenization The SET protocol is obsolete and functionally dead in modern web architecture. Sticking with and hardening your TLS implementation is the only viable path.

Constraint	Why TLS Wins / Why SET Fails
A. High throughput	TLS: Hardware acceleration and session resumption easily handle 10,000+ transactions/minute. SET: Extremely slow due to heavy cryptographic overhead (dual signatures).
B. Browser compatibility	TLS: Natively supported by 100% of modern web browsers. SET: Not supported natively by any modern browser; requires specialized digital wallets.

C. Minimized complexity	TLS: Server-side implementation; entirely transparent to the end-user. SET: Requires distributing digital certificates directly to cardholders, creating a massive logistical nightmare.
D. PCI-DSS compliance	TLS: Required by PCI-DSS (v1.2 or higher). SET: Not recognized as a valid web transport security standard by current PCI bodies.

3. A corporate network is experiencing repeated malware infections that spread rapidly without user interaction. Propose a containment and prevention strategy.

Constraints:

- A. The network consists of 500+ interconnected systems**
- B. Downtime per system should not exceed 10 minutes**
- C. Detection must occur within 2 minutes of infection**
- D. Solution must differentiate between virus, worm, and Trojan behavior**

Problem Context: Rapid, zero-interaction malware spread (Worm behavior) across 500+ systems requiring <10 min downtime, <2 min detection, and behavioral differentiation.

Proposed Solution: Zero Trust Network Access (ZTNA) + AI-Driven EDR Deploy an Endpoint Detection and Response (EDR) solution coupled with strict network micro-segmentation.

- **A. Network of 500+ systems:** Implement micro-segmentation (VLANs/Subnets) to isolate departments. This inherently breaks the lateral movement pathways worms rely on.
- **B. <10 minutes downtime:** Configure the EDR for *Network Containment*. When triggered, the EDR severs the machine's network connection but keeps the OS running. IT can remotely remediate without full system crashes or reboots.
- **C. <2 minutes detection:** Utilize AI/Behavioral EDR (e.g., CrowdStrike, Defender for Endpoint). Instead of waiting for signature scans, these tools detect anomalous API calls and memory injections in milliseconds.
- **D. Differentiate behavior:**
 - *Worm:* Detected via unauthorized lateral port scanning (e.g., SMB/RDP exploitation) without user input.
 - *Virus:* Detected via unauthorized file-modification routines and execution hijacking.
 - *Trojan:* Detected via deceptive execution masking and outbound Command & Control (C2) beaconing to unknown IPs.

4. An organization needs to deploy a firewall architecture to protect its internal network from external threats and insider attacks. Design the solution.

Constraints:

- A. Must support both packet filtering and application-level inspection**
- B. Latency introduced should not exceed 5 ms per request**
- C. Should handle at least 1 Gbps traffic**
- D. Must allow secure remote access for employees**

Problem Context: Protect against internal/external threats, utilize packet filtering and application inspection, maintain <5 ms latency at 1 Gbps+, and allow secure remote access.

Proposed Solution: Hardware-Accelerated Next-Generation Firewall (NGFW) in a DMZ
Deploy a perimeter NGFW that utilizes Single-Pass Parallel Processing (SP3) architecture.

Constraint	How the Solution Addresses It
A. Packet + App inspection	NGFWs natively combine stateful Layer 4 filtering with Layer 7 Deep Packet Inspection (DPI) to identify specific application traffic (e.g., blocking BitTorrent while allowing HTTPS).
B. <5 ms latency	SP3 architecture ensures traffic is opened, inspected for apps, threats, and policies simultaneously in hardware (ASICs), preventing the latency introduced by sequential software scanning.
C. 1 Gbps+ traffic	Modern ASIC-backed firewall appliances are sized specifically for multi-gigabit throughput even with DPI fully engaged.
D. Secure remote access	The NGFW serves as an SSL/IPSec VPN Gateway, enforcing Multi-Factor Authentication (MFA) and ensuring remote workers are inspected by the same security policies as on-premise users.

5. A large enterprise network generates excessive false positives in its IDS, leading to alert fatigue. Propose an optimized IDS/IPS solution.

Constraints:

A. Detection accuracy must be at least 95%

B. False positive rate should be reduced below 5%

C. System must process real-time traffic of 2 Gbps

D. Solution should integrate with existing firewall infrastructure

Problem Context: Excessive false positives in IDS. Needs 95%+ accuracy, <5% false positive rate, 2 Gbps real-time processing, and firewall integration.

Proposed Solution: Hybrid UEBA IPS with Dynamic Firewall Orchestration Transition from purely signature-based rules to User and Entity Behavior Analytics (UEBA) and run the IPS inline with fail-open capabilities.

- **A. 95% Accuracy & B. <5% False Positives:** Accuracy is achieved by layering machine-learning anomaly detection over strict signatures. False positives are drastically reduced by "tuning" the system: whitelisting known-good internal vulnerability scanners and tailoring active signatures strictly to the OS and services actually running on the network.
- **C. 2 Gbps real-time traffic:** Deploy dedicated hardware IPS sensors placed physically inline but configured to "fail-open" (if the sensor crashes, traffic flows unimpeded). Hardware acceleration ensures 2 Gbps is processed without queuing delays.
- **D. Firewall integration:** Implement Security Orchestration, Automation, and Response (SOAR). When the IPS detects a high-confidence threat, it uses APIs to automatically inject a block rule into the edge firewall's dynamic blocklist, cutting off the attacker at the perimeter.